Structural Prediction and Domain Analysis of BRCA1 Using AlphaFold

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Independent Learning Project

Abstract

BRCA1 is an important tumor suppressor protein involved in DNA repair and maintenance of genomic stability. In this study, the predicted three-dimensional structure of the human BRCA1 protein was analyzed using the AlphaFold Protein Structure Database. Structural visualization, pLDDT confidence analysis, Predicted Aligned Error (PAE) analysis, and domain analysis were performed to study the structural organization and flexibility of the protein. The results revealed the presence of highly structured domains along with extensive intrinsically disordered regions within BRCA1. This study demonstrates the application of AlphaFold in structural bioinformatics and provides insight into the structural complexity and functional organization of the BRCA1 protein.

Introduction

Structural bioinformatics helps in understanding the three-dimensional organization and functional properties of proteins using computational tools and biological databases. Protein structure prediction is important because protein structure is closely related to its biological function.

AlphaFold, developed by Google DeepMind, is an artificial intelligence–based tool that predicts highly accurate protein structures from amino acid sequences. The

AlphaFold Protein Structure Database provides predicted models along with confidence metrics such as pLDDT scores and Predicted Aligned Error (PAE).

BRCA1 (Breast Cancer Type 1 Susceptibility Protein) is an important tumor suppressor protein involved in DNA repair, genomic stability, and cell cycle regulation. In this study, the predicted structure of the human BRCA1 protein was analyzed using AlphaFold to study its structural organization, confidence levels, domain arrangement, and flexible regions.

Learning Objective

The objective of this study was to gain practical experience in protein structural analysis using bioinformatics databases and AI-based protein structure prediction tools. This activity helped in understanding the structural organization, domain arrangement, confidence analysis, and functional significance of the BRCA1 protein using AlphaFold. The study also provided hands-on experience in retrieving predicted protein structures, analyzing pLDDT confidence scores and Predicted Aligned Error (PAE) plots, and interpreting structured and intrinsically disordered regions within the protein.

Materials and Methods

The protein structure data used in this study was obtained from the AlphaFold Protein Structure Database. The protein selected for structural analysis was:

- Protein: Breast cancer type 1 susceptibility protein (BRCA1)
- Organism: *Homo sapiens*
- Database: AlphaFold Protein Structure Database and UniProt
- UniProt Accession ID: P38398
- Tool Used: AlphaFold

The predicted three-dimensional structure of BRCA1 was retrieved from the AlphaFold database using the UniProt accession ID. Structural visualization, confidence analysis using pLDDT scores, Predicted Aligned Error (PAE) analysis, and domain organization analysis were performed to study the structural features and flexibility of the protein.

Workflow of the Analysis

1. Search for the BRCA1 protein in the AlphaFold Protein Structure Database
2. Retrieve the predicted BRCA1 structure using the UniProt accession ID (P38398)
3. Visualize the three-dimensional structure of the protein
4. Analyze pLDDT confidence scores and confidence-colored regions
5. Examine the Predicted Aligned Error (PAE) plot
6. Identify structured domains and intrinsically disordered regions
7. Download structure files and document observations and screenshots

Results

The structural analysis of the BRCA1 protein using the AlphaFold Protein Structure Database provided important insights into its three-dimensional organization, structural confidence, and domain architecture. The predicted model of the human BRCA1 protein revealed a complex structure containing both highly ordered domains and extensive intrinsically disordered regions.

The confidence-colored structural visualization showed that only specific regions of the protein possessed high pLDDT confidence scores, represented by blue-colored domains, indicating stable and reliable structural conformations. In contrast, large portions of the protein appeared in yellow, orange, and red colors, suggesting low-confidence and highly flexible regions.

The Predicted Aligned Error (PAE) analysis further demonstrated high confidence within individual structured domains while showing increased uncertainty between distant regions of the protein. This indicates that BRCA1 contains multiple independently folded domains connected through flexible linker regions.

Domain analysis revealed the presence of compact structural regions spatially separated by long disordered segments, highlighting the modular organization of the BRCA1 protein. Rotational visualization of the structure further emphasized the extended and flexible architecture of the protein.

Overall, the AlphaFold prediction provided valuable insights into the structural complexity, flexibility, and functional organization of BRCA1, demonstrating the usefulness of AI-based protein structure prediction tools in structural bioinformatics.

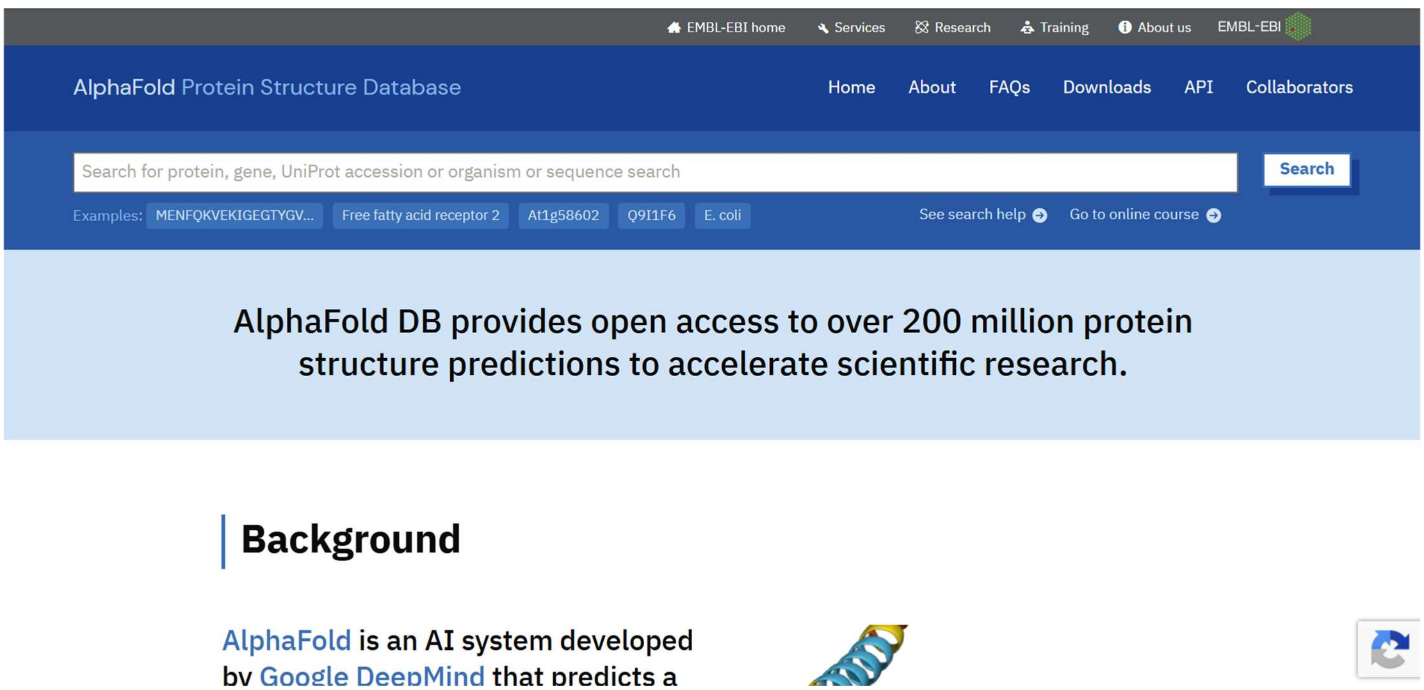

Figure 1: Homepage of the AlphaFold Protein Structure Database showing search and navigation tools for predicted protein structures.

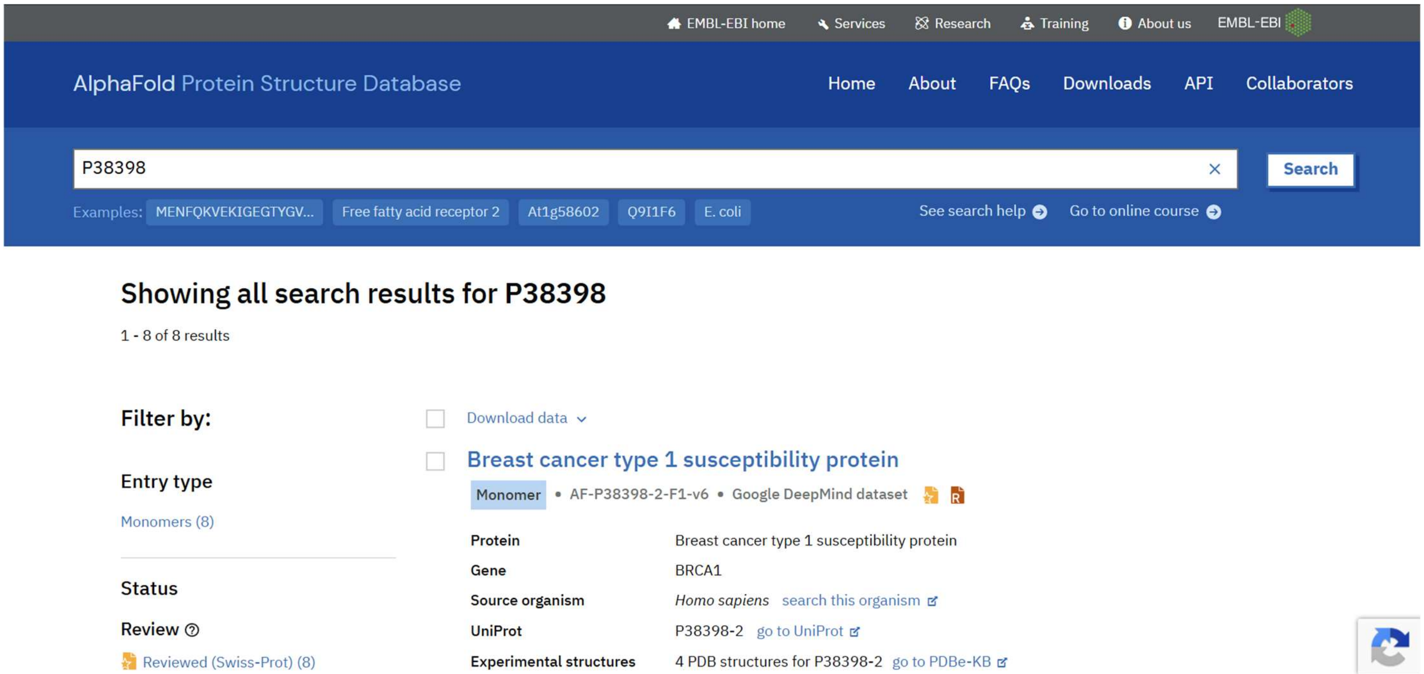

Figure 2: Search results for the BRCA1 protein in the AlphaFold database showing multiple related protein entries.

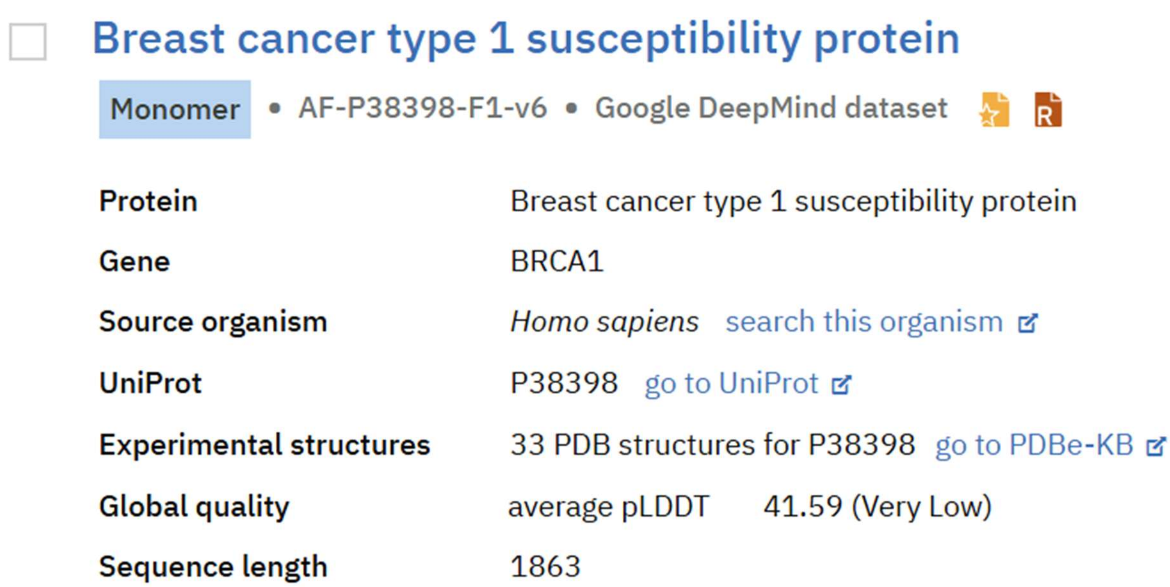

Figure 3: Canonical AlphaFold entry for the human BRCA1 protein (UniProt ID: P38398) displaying protein information and structural confidence details.

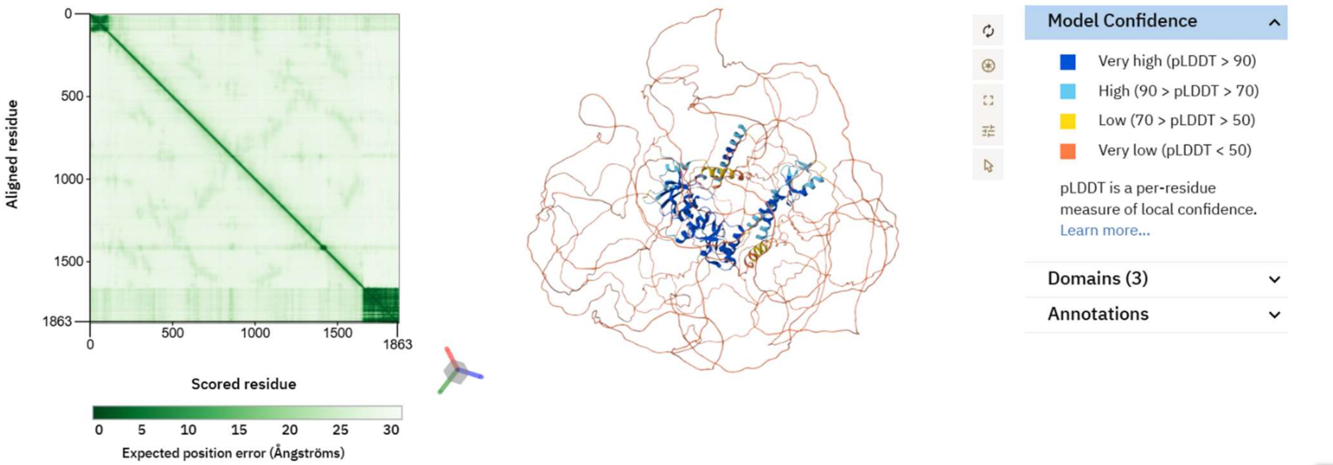

Figure 4: AlphaFold-predicted three-dimensional structure of BRCA1 along with the Predicted Aligned Error (PAE) plot and pLDDT confidence visualization.

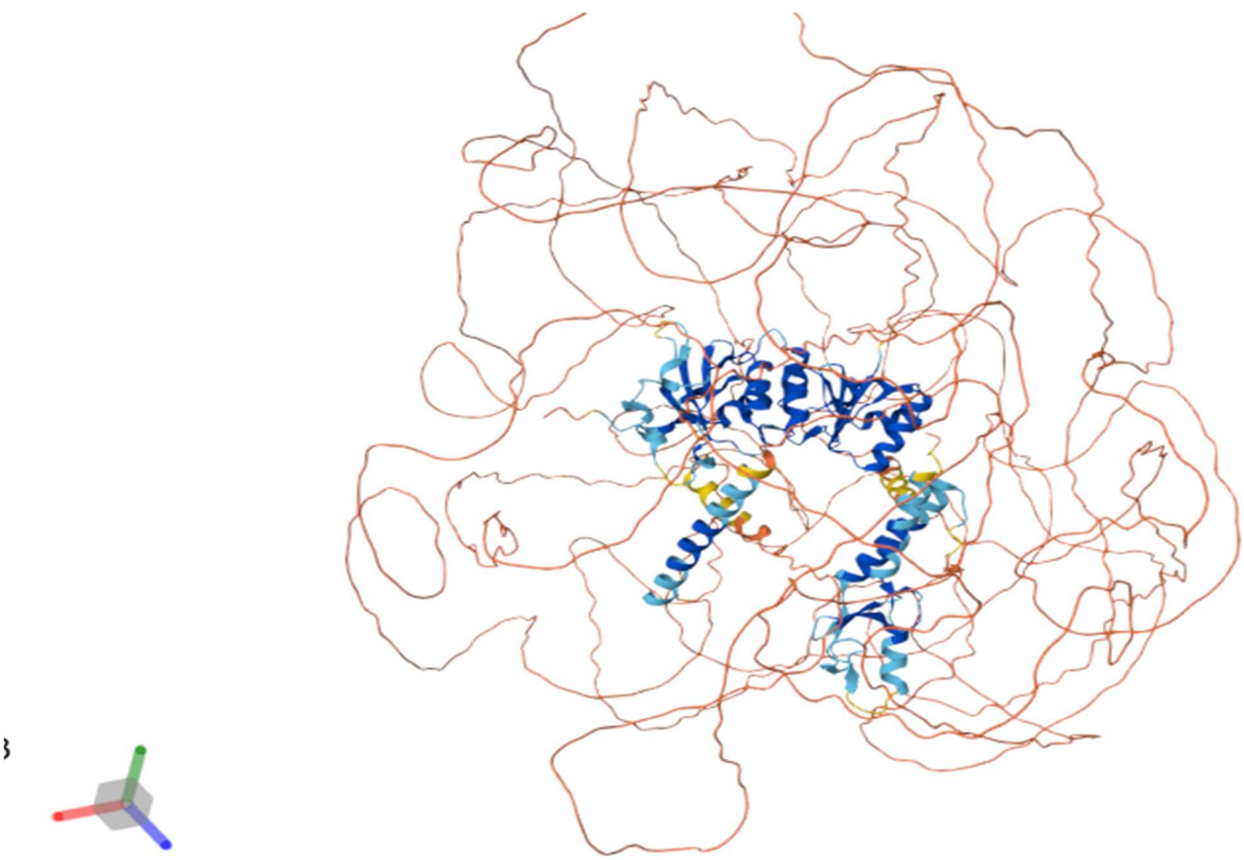

Figure 5: Confidence-colored structural visualization of BRCA1 showing highly structured domains and intrinsically disordered regions.

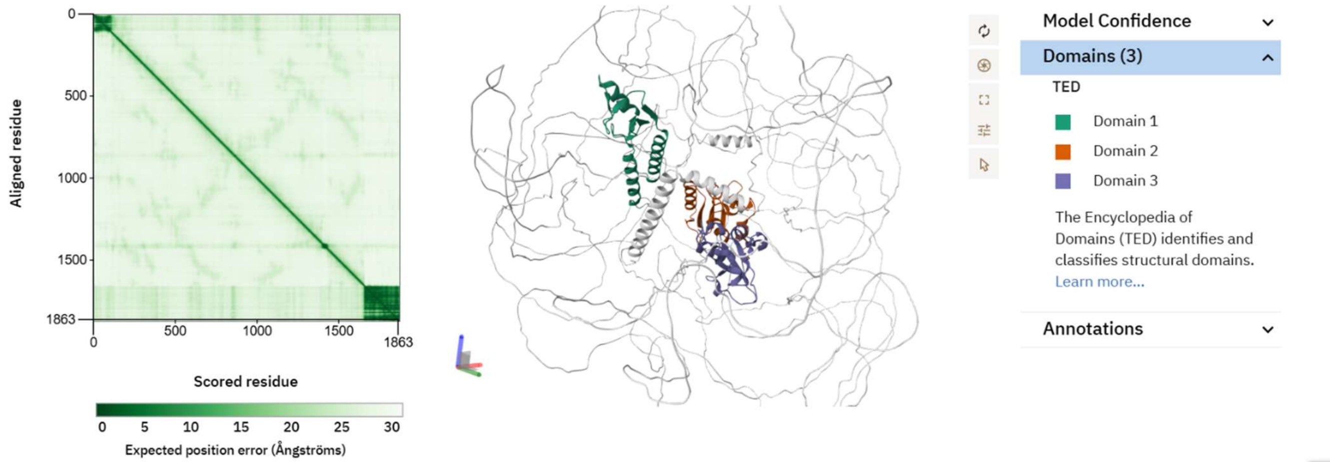

Figure 6: Predicted Aligned Error (PAE) analysis and domain segmentation view of the BRCA1 protein.

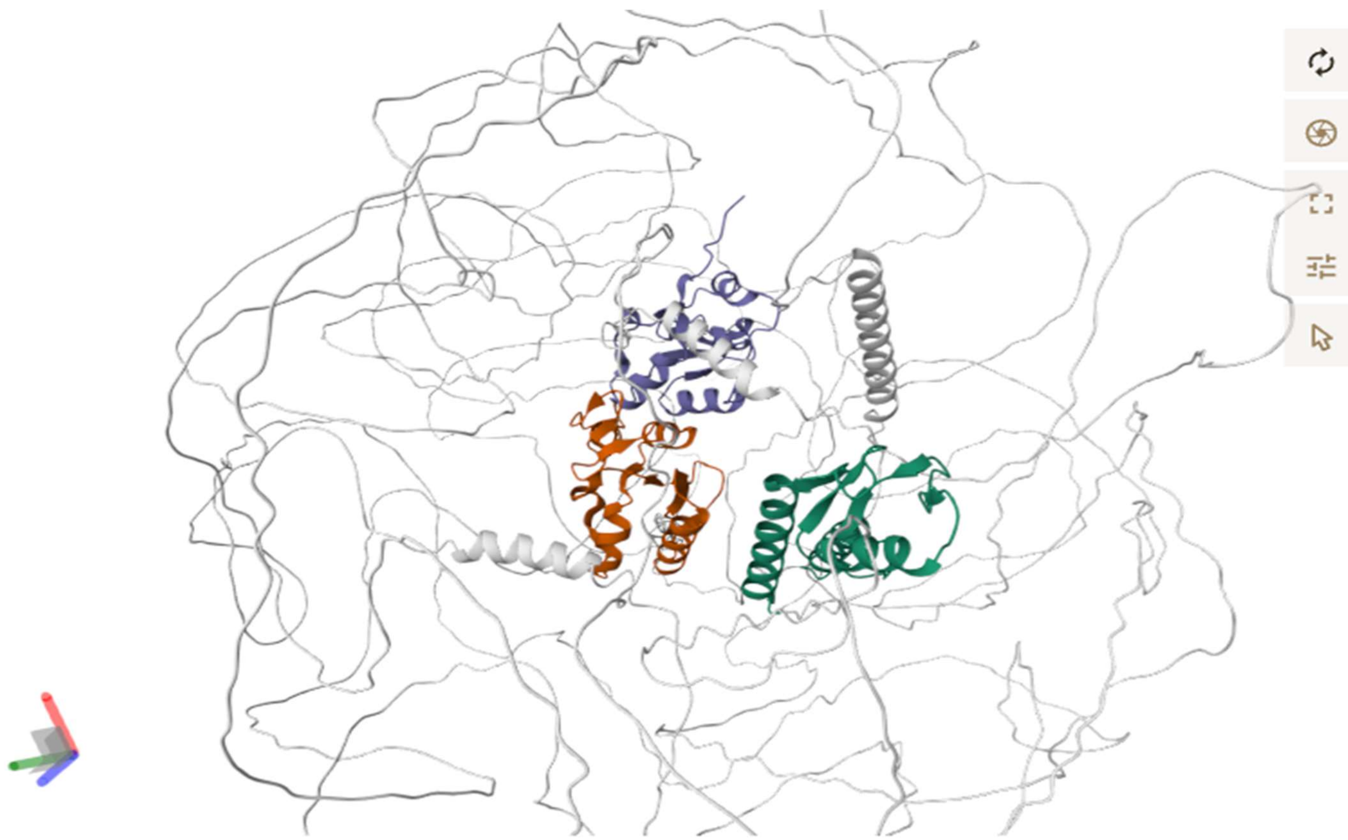

Figure 7: Domain analysis of the BRCA1 protein showing compact structured domains separated by flexible linker regions.

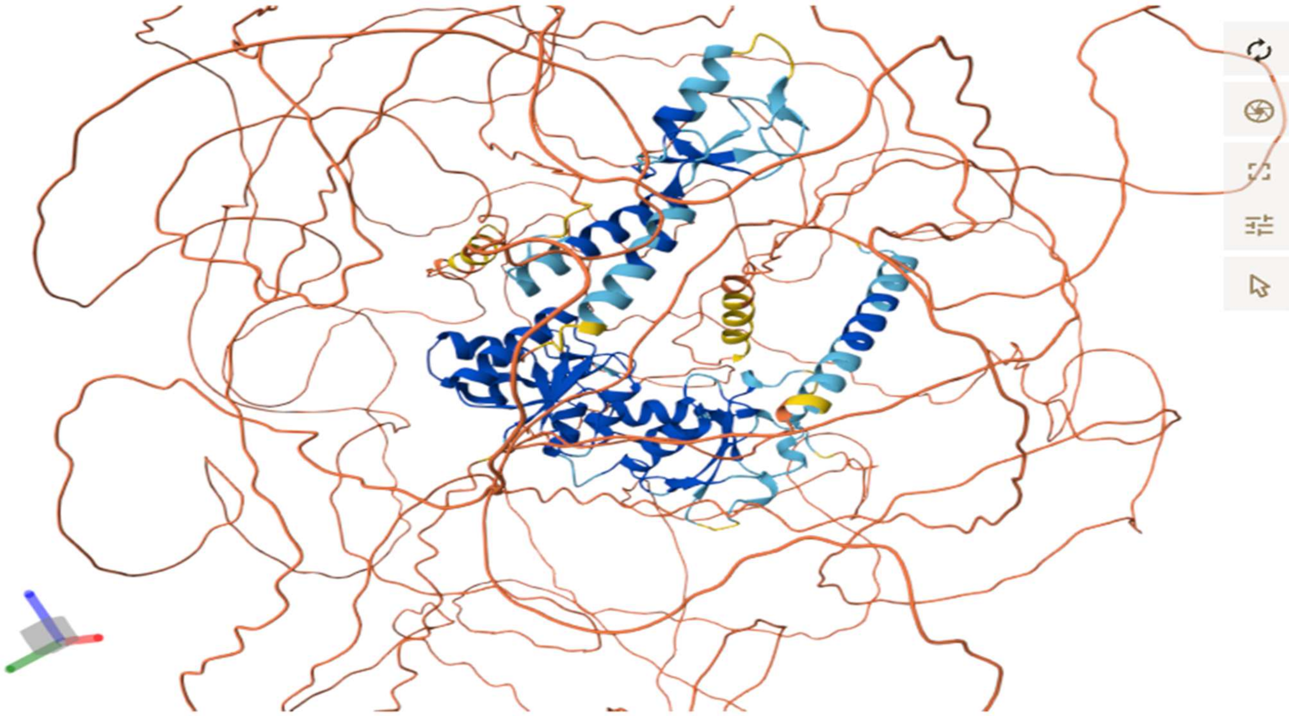

Figure 8: Side-view visualization of the BRCA1 protein demonstrating the spatial arrangement of structured and disordered regions.

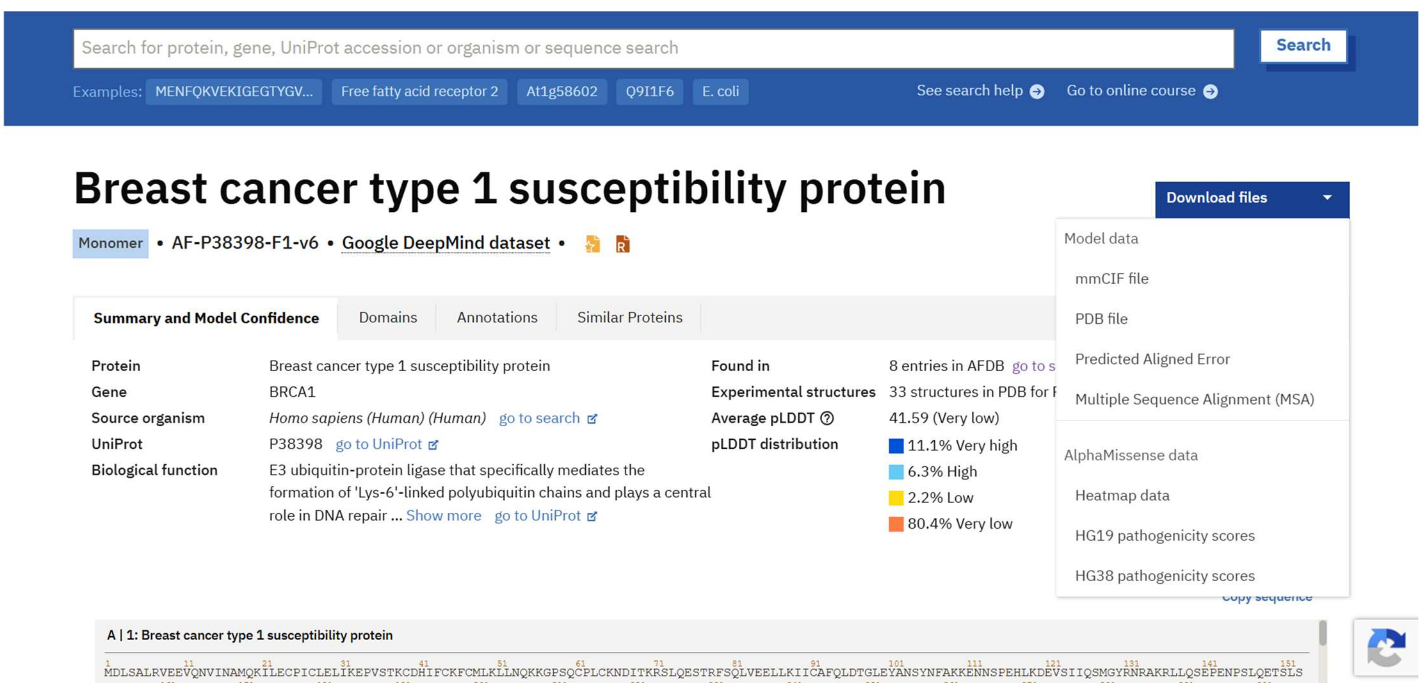

Figure 9: AlphaFold structure download section showing export options for PDB and mmCIF files.

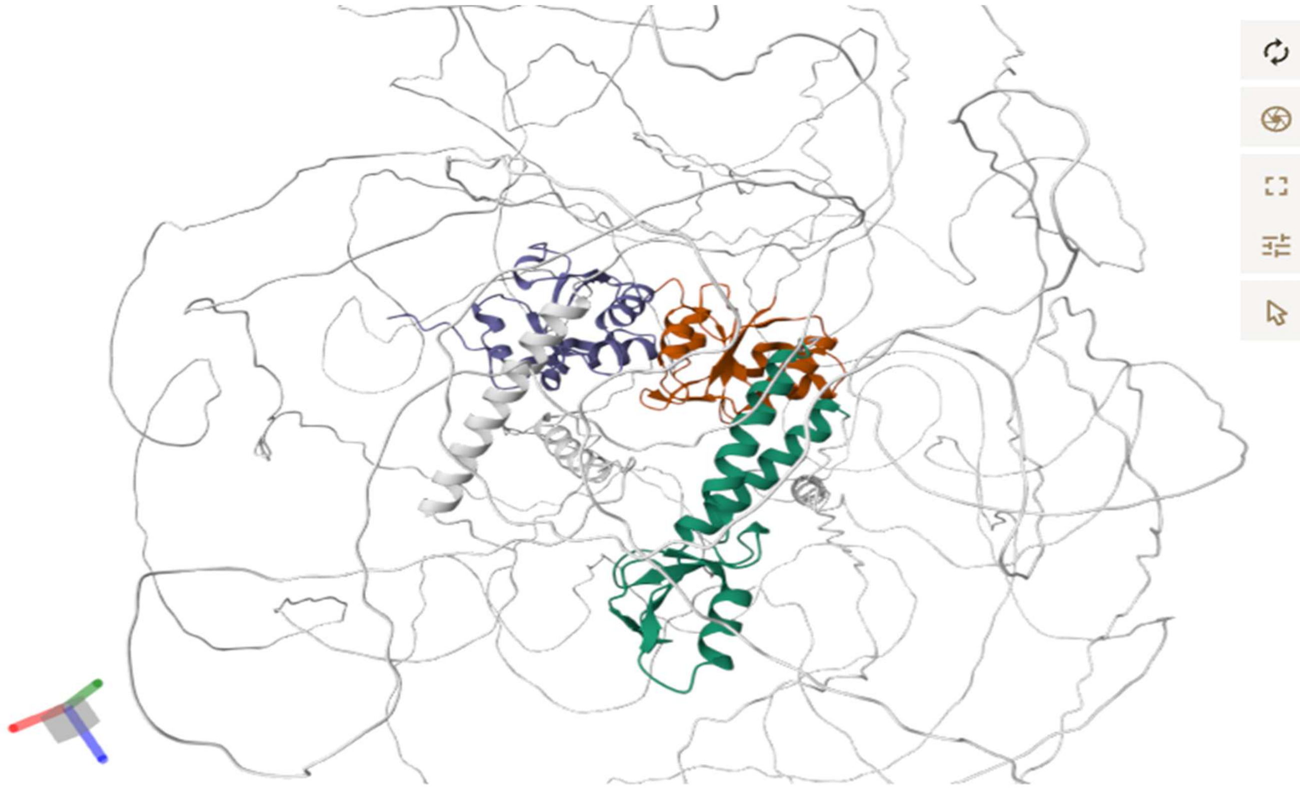

Figure 10: Domain-oriented structural visualization of BRCA1 showing spatial separation of predicted structural domains.

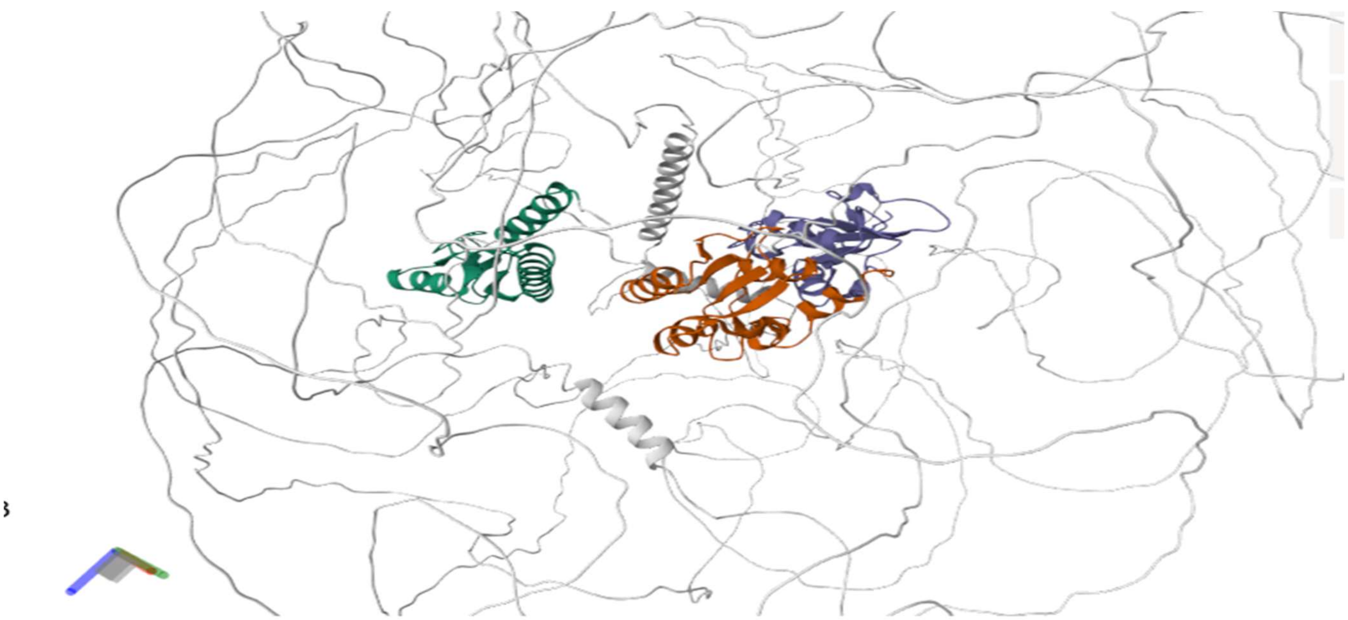

Figure 11: Final structural visualization of BRCA1 summarizing the predicted architecture and domain organization of the protein.

Discussion

The AlphaFold analysis of the BRCA1 protein provided important insights into its structural organization and functional significance. The predicted structure revealed the presence of both highly structured domains and extensive intrinsically disordered regions within the protein.

The confidence-colored visualization and pLDDT analysis showed that only certain regions of BRCA1 possess highly reliable structural conformations, while several regions displayed low confidence scores, indicating structural flexibility and disorder. The Predicted Aligned Error (PAE) analysis further demonstrated strong confidence within individual domains but greater uncertainty between distant regions of the protein.

Domain analysis highlighted the modular organization of BRCA1, with compact structured regions separated by flexible linker segments. These structural features are closely related to the biological role of BRCA1 in DNA repair and maintenance of genomic stability. Overall, the study demonstrates the usefulness of AlphaFold in understanding protein structure, flexibility, and functional organization through computational bioinformatics approaches.

Limitations of the Study

Although the structural analysis of the BRCA1 protein using AlphaFold provided valuable insights, certain limitations must be considered.

Firstly, the study was entirely based on computational predictions and publicly available database information without experimental validation. The accuracy of the analysis depends on the reliability of the predicted structural model generated by AlphaFold.

Secondly, several regions of the BRCA1 protein showed low confidence scores, indicating the presence of intrinsically disordered or highly flexible regions that are difficult to predict accurately through computational methods alone.

Thirdly, the study focused mainly on structural visualization and confidence analysis. Advanced analyses such as molecular docking, mutation impact studies, molecular dynamics simulations, and protein–protein interaction analysis were not included.

Despite these limitations, the study provides a clear understanding of the application of AI-based protein structure prediction tools in structural bioinformatics and protein analysis.

Future Scope

The present study provides a basic understanding of protein structural analysis using AlphaFold. However, several advanced analyses can be carried out in the future to obtain deeper insights into the structure and function of the BRCA1 protein.

Future studies may include detailed structural visualization using software such as PyMOL and Discovery Studio to analyze protein domains, molecular surfaces, and interaction regions. Comparative structural analysis with experimentally determined structures from the Protein Data Bank can also be performed to evaluate structural similarities and differences.

Further research may involve mutation analysis, protein–protein interaction studies, molecular docking, and computational modeling to better understand the role of BRCA1 in DNA repair pathways and cancer development. Comparative structural studies across multiple species may also provide additional insights into the evolutionary conservation and biological significance of the BRCA1 protein.

Overall, advanced bioinformatics and structural biology tools can provide a more comprehensive understanding of the architecture, flexibility, and functional importance of BRCA1.

Conclusion

This study demonstrates the application of AlphaFold in protein structural analysis and computational bioinformatics. In this project, the predicted three-dimensional structure of the human BRCA1 protein was analyzed to study its structural organization, confidence levels, and domain arrangement.

The AlphaFold prediction revealed that BRCA1 contains both highly structured domains and extensive intrinsically disordered regions, indicating the complex and flexible nature of the protein. Confidence analysis using pLDDT scores and Predicted Aligned Error (PAE) plots provided important insights into the reliability and spatial organization of the predicted structure.

This study provided practical experience in protein structure visualization and interpretation using AI-based bioinformatics tools. Overall, the project highlights the importance of computational approaches in understanding protein architecture, structural flexibility, and biological significance.

References

- AlphaFold Protein Structure Database
- UniProt Consortium
- Jumper J et al. Highly accurate protein structure prediction with AlphaFold
- National Center for Biotechnology Information (NCBI)
- Protein Data Bank (PDB)
- DeepMind AlphaFold Resources